

AI-POWERED MUSEUM NAVIGATION FOR THE VISUALLY IMPAIRED

**ML on Multimedia Data –
MSc in AI**

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INSPIRATION

accessibility

In a gallery, the visitor's experience is built on the emotions sparked by seeing an artwork – its colors, forms, and composition. For people who are visually impaired, this experience must be perceived through other senses. Traditional aids like audio guides require manual preparation for each artwork and do not scale. This inspired our goal: **an automatic system that turns any painting into an expressive spoken description, making art accessible through sound.**



THEORETICAL BACKGROUN

- Transformers introduced the self-attention mechanism, becoming the foundation of modern deep learning models for sequences.
- The Vision Transformer (ViT) extended this idea to images, by treating image patches as a sequence – enabling a single architecture for both text and images.
- This led to Vision-Language Models (VLMs), which combine a vision encoder and a text decoder to generate natural-language descriptions of images (image captioning).
- Text-to-Speech (TTS) then converts the generated text into natural-sounding speech, using neural synthesis with prosody control (pitch, rate, volume) via SSML.



TRAINING DATASET



2,930 paintings
(13th–21st century)

ARTPEDIA DATASET

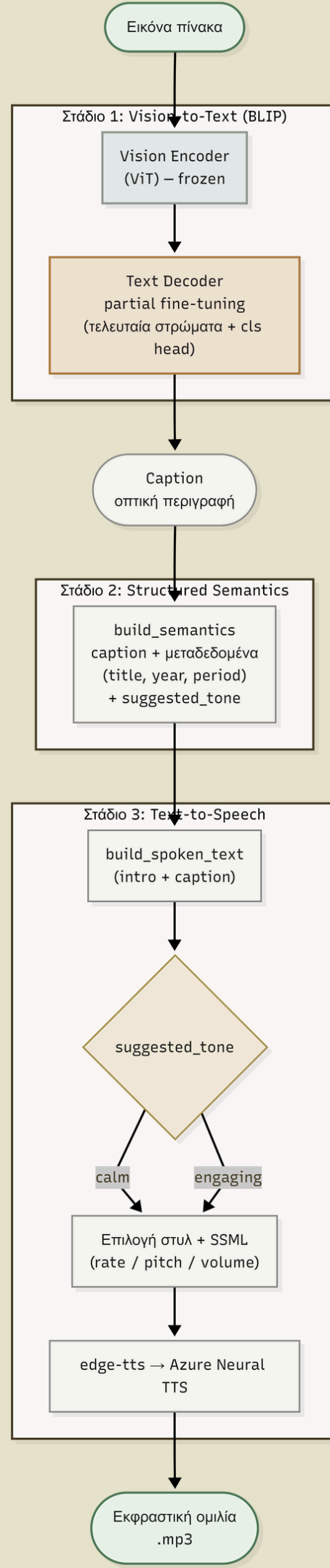


visual and contextual
sentences

DATA PREPARATION

- Artpedia – 2,930 paintings (13th–21st century), each with visual and contextual sentences.
- Used only visual sentences → the model describes what is seen, not unseen facts (artist, history).
- Rule-based cleaning removed mislabeled contextual sentences (attribution, provenance, dates) – ~8% of sentences.
- Final dataset: 1,935 train / 286 validation / 296 test.





ARCHITECTURE overview

A three-stage multimodal pipeline:

- 1. **Vision-to-Text** – painting → visual caption (partial fine-tuned BLIP)
- 2. **Structured Semantics** – caption + metadata (title, period) + narration tone
- 3. **Text-to-Speech** – expressive spoken description



VISION-TO-TEXT

Model

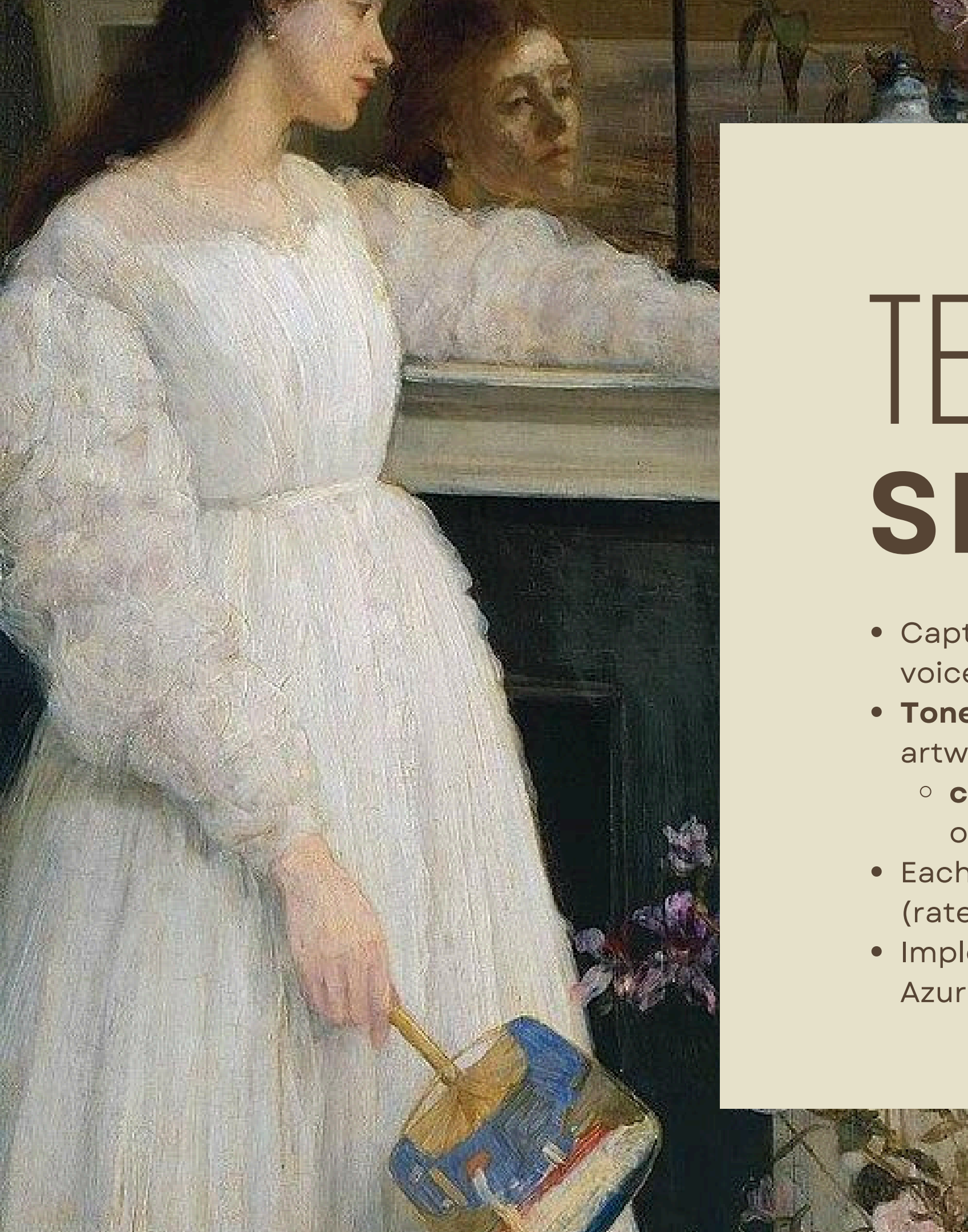
- Base: pre-trained BLIP (blip_caption), via LAVIS – vision encoder (ViT) + text decoder.

Partial fine-tuning strategy

- Vision encoder: frozen
- Text decoder: only the last layer (layer 11) + word-prediction head
- 10.1M / 224M parameters trained (~4.5%) → limits overfitting on small data

Training hyperparameters

- Optimizer: AdamW · Learning rate: 1×10^{-5}
- Batch size: 4 · Max epochs: 20
- Mixed precision: fp16
- Early stopping (patience 2) on validation loss

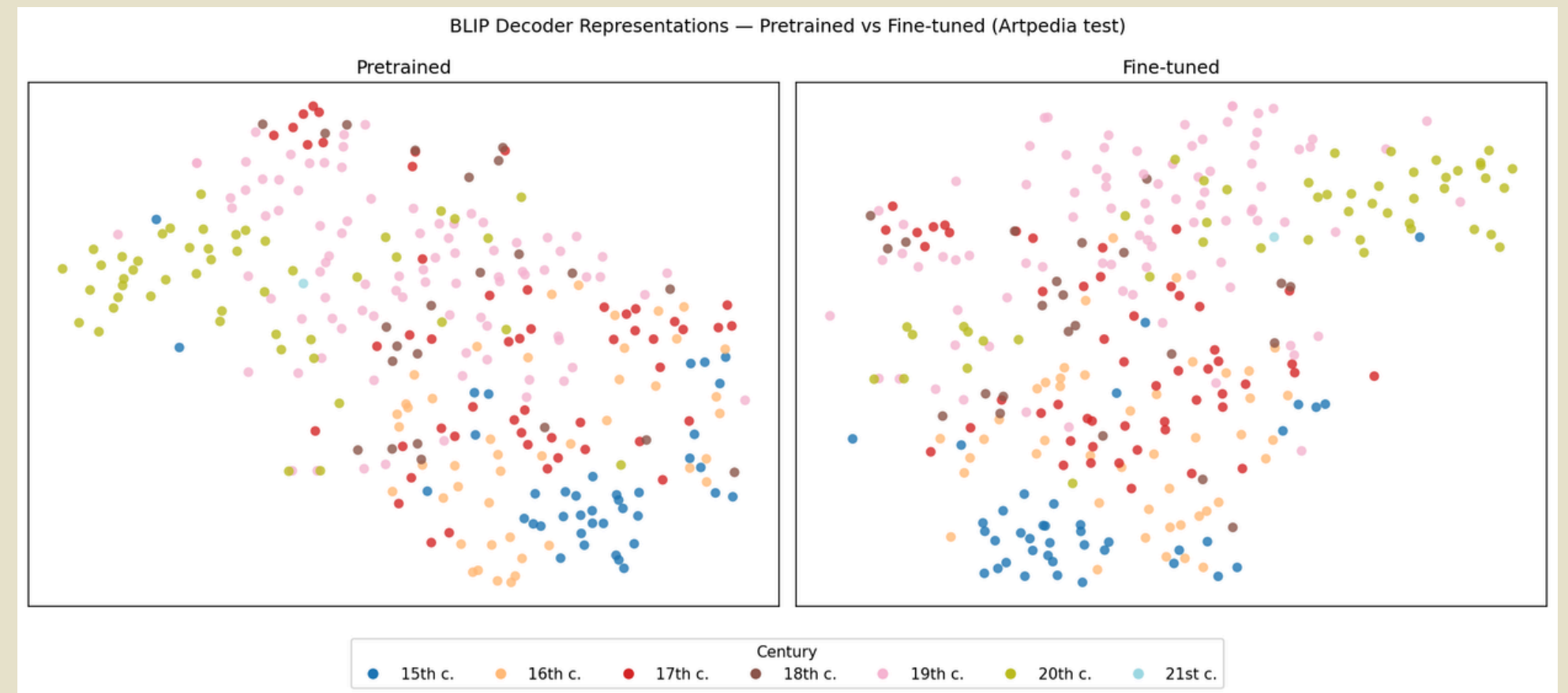


TEXT-TO-SPEECH

- Caption enriched with **metadata**, then voiced as speech.
- **Tone chosen automatically** from the artwork's date:
 - **calm** (before 1800) · **engaging** (1800 onwards)
- Each tone → distinct voice + prosody (rate, pitch, volume).
- Implemented with **edge-tts** (Microsoft Azure Neural TTS) via SSML.

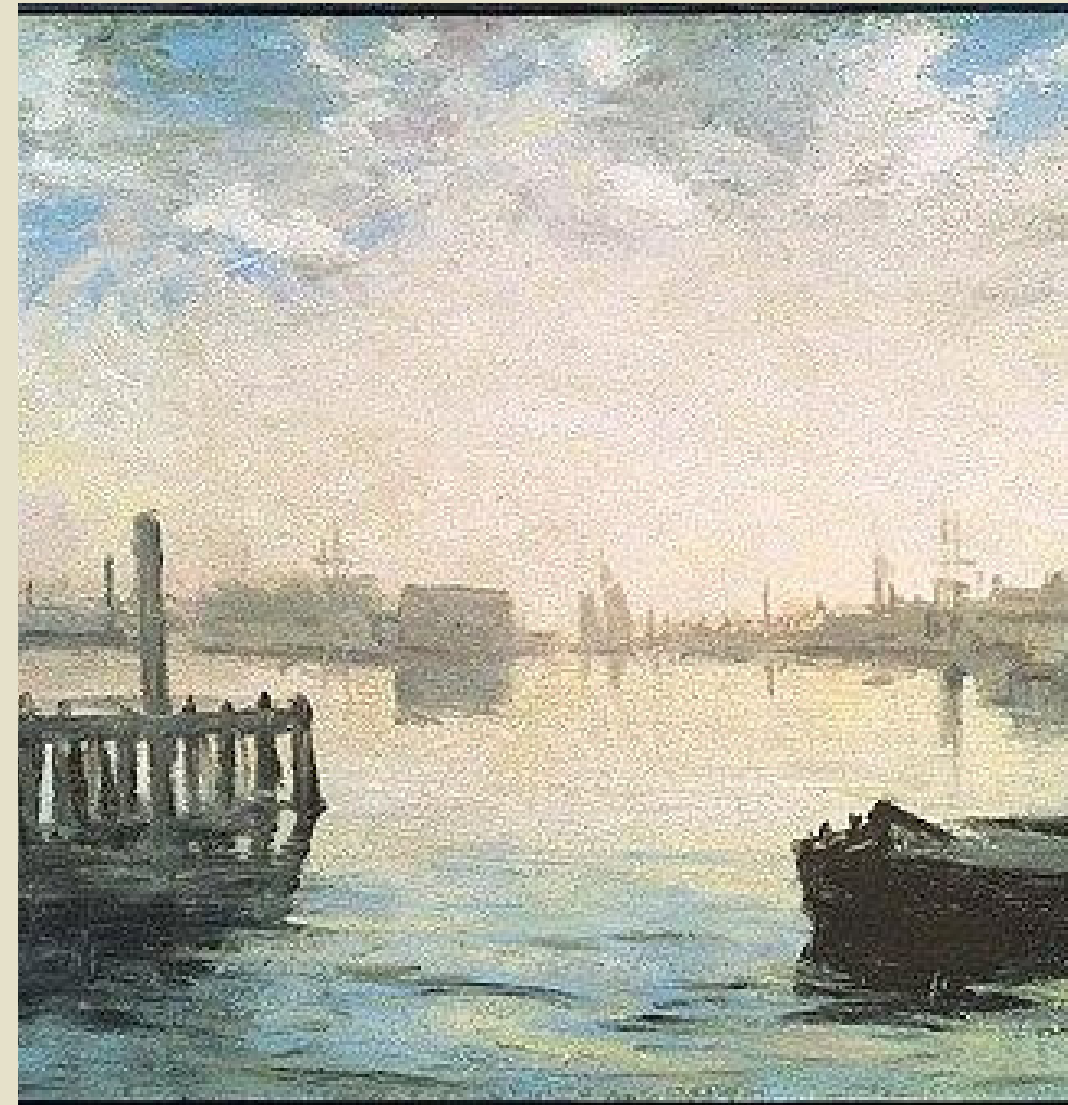
RESULTS — QUANTITATIVE

Model	BLEU-4	CIDEr
Pretrained	0.013	0.027
Fine-tuned	0.034	0.062



ANNOTATION

Done in oil on canvas, the impressionist painting depicts morning in the harbor of Gloucester.



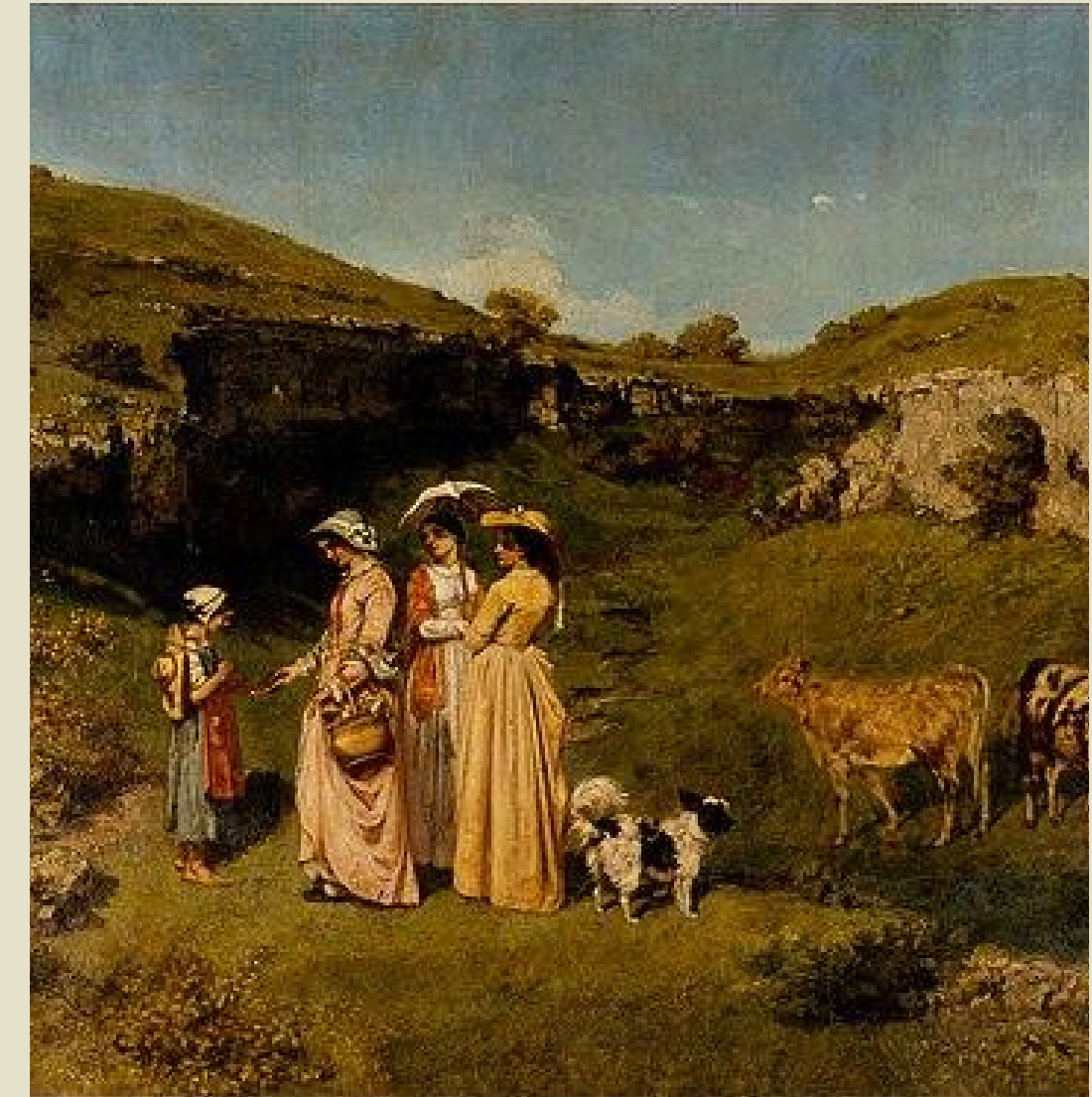
GLOUCESTER HARBOR (1877)

PRETRAINED A painting of boats in the water near a dock.

FINE-TUNED The painting depicts the harbor at dawn, with the docks in the foreground.

ANNOTATION

In the centre three young women offer a barefoot peasant girl something to eat. One holds an umbrella against the sun. A dog stands behind the trio, whilst to the right two cows graze in a hilly landscape.



YOUNG LADIES OF THE VILLAGE, 1852

A painting of people and animals in a field.

The painting depicts three women in a field with cows, one is holding an umbrella and the other is standing on a hill.

RESULTS — ZERO-SHOT ROBUSTNESS

The VizWiz dataset

- Photos taken by blind people using mobile phones – the first captioning dataset for this real use case.
- Images are often blurry, poorly framed, or badly lit – reflecting real capture conditions.
- We evaluate zero-shot (no training on VizWiz) on 200 validation images.
- On real-world blind-user photos, the fine-tuned model performs worse.
- Reveals the **specialization vs. generalization trade-off**.



Model	BLEU-4	CIDEr
Pretrained	0.176	0.298
Fine-tuned	0.117	0.265

CALM

voice: en-GB-SoniaNeural
rate : -5% , pitch: -2Hz, volume +0%

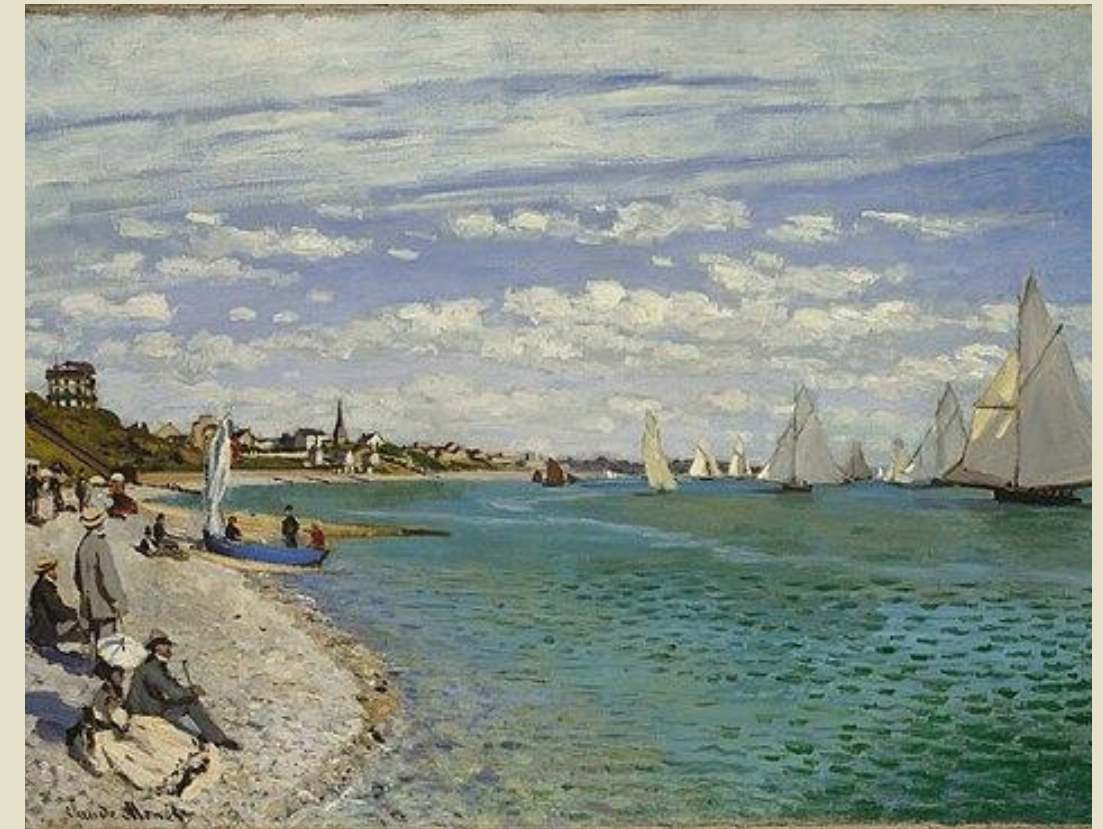


LANDSCAPE WITH WATERFALL, 1660

The painting depicts a river in a wooded landscape, with a fallen tree in the foreground.

ENGAGING

voice: en-US-JennyNeural
rate : -2% , pitch: +4Hz, volume +3%



REGATTA AT SAINTE-ADRESSE, 1867

Done in oil on canvas, the painting depicts a view of the beach at Trouville.



CONCLUSIONS

- **Built an end-to-end pipeline:**
 - painting → description → expressive speech.
- **Domain adaptation works:**
 - fine-tuning on Artpedia more than doubled BLEU-4 and CIDEr, producing richer art captions.
- **Specialization has a cost:**
 - the fine-tuned model lost robustness on real-world VizWiz photos – a clear specialization vs. generalization trade-off.
- **Honest by design:**
 - the caption describes only what is seen; metadata and tone are added separately, so the system works even on unknown artworks.



FUTURE Work

- **The gap:** Artpedia images are clean reproductions – unlike the noisy photos a blind user would actually take.
- **Proposed solution:** use a diffusion model to generate synthetic paintings with VizWiz-like noise (blur, framing, lighting). Augment Artpedia with this synthetic data → train on a distribution closer to real use.
- **Goal:** bridge the specialization vs. generalization gap identified in our evaluation.



THANK YOU
questions?